

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1518

South Colton 138 kV
September 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1518. The Interconnection Customer proposes a 303.6 MW battery energy storage facility interconnecting at the South Colton 138 kV substation in the Virginia area. The request was received on July 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 357 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,404,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Grassmere – Red Hill 138.0 kV
Contingency: Clover – Bear 345.0 kV line
Loading: 114.5% (Rating: 295 MVA)
DFAX: 5.6%
Pre-project loading: 108.7%
2. Facility: Coal Mtn – Grassmere 138.0 kV line
Contingency: Bear – Clover 345.0 kV
Loading: 113.1% (Rating: 329 MVA)
DFAX: 5.6%
Pre-project loading: 107.9%
3. Facility: Coal Mtn – Central Austin 138.0 kV
Contingency: Bear – Clover 345.0 kV
Loading: 112.7% (Rating: 338 MVA)
DFAX: 5.6%
Pre-project loading: 107.7%
4. Facility: Waynesboro – Central Austin 138.0 kV line
Contingency: Bear – Clover 345.0 kV
Loading: 112.4% (Rating: 348 MVA)
DFAX: 5.6%
Pre-project loading: 107.5%
5. Facility: Alderwood Substation – Waynesboro 138.0 kV
Contingency: Bear – Clover 345.0 kV
Loading: 111.5% (Rating: 377 MVA)
DFAX: 5.6%
Pre-project loading: 107.0%

6. Facility: Monel – Candelaria 138.0 kV
Contingency: Bear – Clover 345.0 kV
Loading: 107.0% (Rating: 443 MVA)
DFAX: 8.2%
Pre-project loading: 101.3%
7. Facility: Glen Oaks – S Texa 138.0 kV
Contingency: Brush Tavern – Rustburg 138.0 kV line
Loading: 106.5% (Rating: 173 MVA)
DFAX: 14.7%
Pre-project loading: 80.7%
8. Facility: Yaphank – S Colton 138.0 kV
Contingency: Bear – Clover 345.0 kV L
Loading: 106.3% (Rating: 401 MVA)
DFAX: 17.6%
Pre-project loading: 93.0%
9. Facility: Evington Substation – Canby 138.0 kV L
Contingency: Brush Tavern – Rustburg 138.0 kV
Loading: 105.8% (Rating: 229 MVA)
DFAX: 14.7%
Pre-project loading: 86.4%
10. Facility: Weston – Glen Oaks 138.0 kV
Contingency: Brush Tavern – Rustburg 138.0 kV
Loading: 103.2% (Rating: 196 MVA)
DFAX: 14.7%
Pre-project loading: 80.4%
11. Facility: Canby – Weston 138.0 kV L
Contingency: Rustburg – Brush Tavern 138.0 kV
Loading: 102.2% (Rating: 215 MVA)
DFAX: 14.7%
Pre-project loading: 81.4%
12. Facility: Bear – Mecklenburg Power Sta 345.0 kV line
Contingency: Bear – Clover 345.0 kV
Loading: 101.8% (Rating: 2,827 MVA)
DFAX: 19.5%
Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grassmere – Red Hill 138.0 kV	Rebuild line with higher-capacity conductor	\$3,996,000	30% (\$1,186,000)
Coal Mtn – Grassmere 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,810,000	30% (\$1,131,000)
Central Austin – Coal Mtn 138.0 kV	Rebuild line with higher-capacity conductor	\$5,960,000	30% (\$1,769,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Central Austin – Waynesboro 138.0 kV	Rebuild line with higher-capacity conductor	\$3,923,000	30% (\$1,165,000)
Alderwood Substation – Waynesboro 138.0 kV	Rebuild line with higher-capacity conductor	\$3,770,000	30% (\$1,119,000)
Monel – Candelaria 138.0 kV	Reconductor line	\$14,914,000	20% (\$3,002,000)
Glen Oaks – South Texa 138.0 kV	Reconductor line	\$1,051,000	30% (\$320,000)
Yaphank – S Colton 138.0 kV	Reconductor line	\$1,704,000	30% (\$512,000)
Evington Substation – Canby 138.0 kV L	Reconductor line	\$6,879,000	30% (\$2,092,000)
Glen Oaks – Weston 138.0 kV	Reconductor line	\$1,419,000	30% (\$432,000)
Canby – Weston 138.0 kV	Reconductor line	\$9,101,000	30% (\$2,768,000)
Bear – Mecklenburg Power Station 345.0 kV	Reconductor line	\$52,475,000	4% (\$1,908,000)
	Total allocated to project		\$17,404,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.